Practice workbook · Calculating the mean

Finding or describing the arithmetic mean of a data set.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: PSY-RM-15

1. Question 2 · part c · 1 mark

2025 JUN · Paper 02

Worked-example retry · Attempt part c only.

2 Jenna investigated whether stress affected performance on a memory test.

She recruited 10 participants from her university. Participants were asked to remember a list of 20 words. Fifteen minutes later, they were asked to write down as many words as they could remember. The same participants were tested twice:

- Test 1: one week before their examinations began.
- Test 2: one week after their examinations had finished.

(a) Describe how Jenna could use an opportunity sampling method to gather the participants for her investigation.

(2)

(b) State which experimental research design Jenna used in her investigation.

(1)

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1. Question 2 · part c · 1 mark

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Worked-example retry · Attempt part c only.

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Jenna recorded the total number of correct words remembered by participants both before and after their examinations. Her results are shown in **Table 2**.

Participant	Total number of correct words remembered one week before examinations began (out of 20)	Total number of correct words remembered one week after examinations finished (out of 20)
A	5	8
B	6	7
C	7	9
D	4	8
E	5	7
F	3	7
G	9	9
H	10	11
I	6	6
J	5	8

Table 2

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(c) Calculate the mean score for the total number of correct words remembered one week **before** the participants' examinations began.

(1)

Mean .....

(d) Calculate the median score for the total number of correct words remembered one week **after** the participants' examinations finished.

(2)

Median .....

(Total for Question 2 = 6 marks)

1. Question 2 · part c · 1 mark · Working and answer

Working and answer

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2. Question 2 · part a · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

2 Michelle conducted a study to find out if 15 year olds had better long-term memory recall than 55 year olds. She gave two groups of participants one hour to memorise a set of 40 pictures. After two weeks, she asked the participants to recall as many of the pictures as they could.

Her results are shown in **Table 1**.

15 year olds	Number of pictures recalled correctly	55 year olds	Number of pictures recalled correctly
Participant A	35	Participant A	16
Participant B	30	Participant B	28
Participant C	29	Participant C	27
Participant D	18	Participant D	34
Participant E	38	Participant E	35
Participant F	31	Participant F	26
Participant G	24	Participant G	21

Table 1

(a) Calculate the mean score for the number of pictures recalled by 55 year olds.

You must give your answer to two decimal places.

(2)

Mean score for 55 year olds .....

(b) Calculate the median score for the number of pictures recalled by 15 year olds.

(2)

Median score for 15 year olds .....

(c) Calculate the range for the number of pictures recalled by all participants.

(1)

Range for all participants .....

(Total for Question 2 = 5 marks)

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2. Question 2 · part a · 2 marks · Working and answer

Working and answer

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3. Question 3 · part a · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part a only.

3 Ryan used a questionnaire to assess whether students had a fixed or growth mindset in four secondary schools. He asked the students to complete the questionnaire online.

His results are shown in **Table 2**.

School	Number of students with a fixed mindset	Number of students with a growth mindset
A	458	931
B	211	439
C	641	838
D	189	271

Table 2

(a) Calculate the mean number of students with a fixed mindset.

(1)

Mean .....

(b) Calculate the total number of students who completed the questionnaire.

(1)

Total .....

(c) Calculate the range for the number of students with a growth mindset.

(1)

Range .....

(Total for Question 3 = 3 marks)

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3. Question 3 · part a · 1 mark · Working and answer

Working and answer

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4. Question 3 · part d · 1 mark

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Further practice · Attempt part d only.

3 Quentin is investigating the capacity of short-term memory. He showed 20 objects to 10 participants for 30 seconds. Quentin then asked them to recall as many as they could.

His results are shown in **Table 2**.

Participant	Number of objects recalled
A	10
B	9
C	8
D	8
E	6
F	7
G	8
H	9
I	10
J	11

Table 2

(a) Calculate how many participants recalled nine objects as a fraction of all participants.

You must give your answer in the lowest form.

(1)

Fraction .....

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4. Question 3 · part d · 1 mark

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Further practice · Attempt part d only.

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(b) Calculate the median for the number of objects recalled.

(2)

Median .....

(c) Calculate the mode for the number of objects recalled.

(1)

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Mode .....

(d) State how Quentin would calculate the mean score for the number of objects recalled.

(1)

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(Total for Question 3 = 5 marks)

4. Question 3 · part d · 1 mark · Working and answer

Working and answer

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